Alexandria University
Faculty of Engineering
Computer and Systems Engineering Department

CSE343 Embedded Systems
Assigned: Sunday 12/04/2026
Due: Sunday 19/04/2026

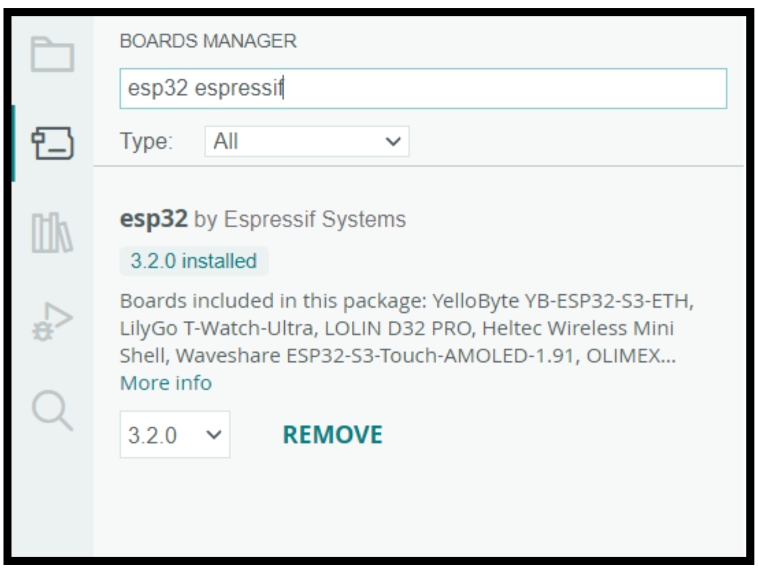
Lab Experiment 05: Arduino IoT Cloud

Problem Statement

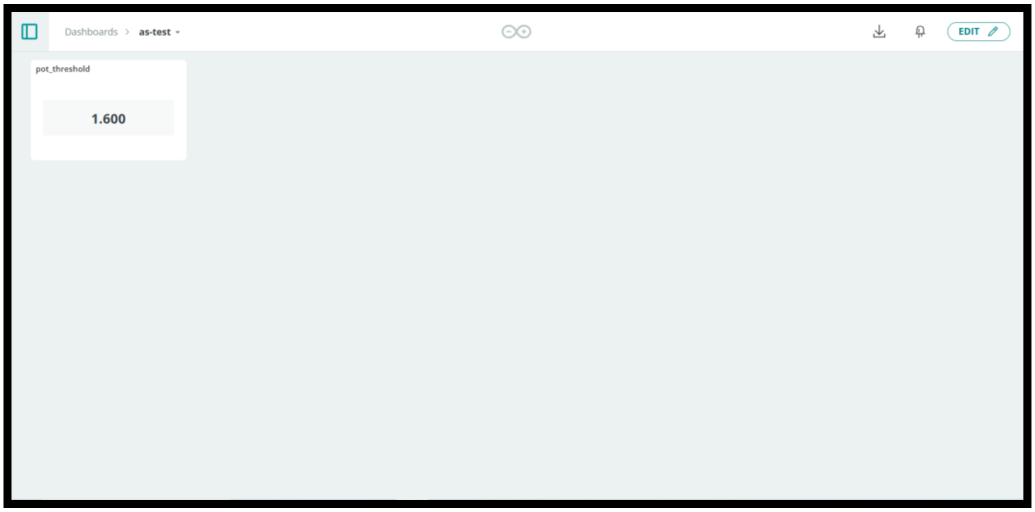
You are required to design and implement an IoT-based system using the Arduino IoT Cloud platform, an ESP32 microcontroller, a potentiometer, and an LED. The system should allow a user to remotely set a voltage threshold via the cloud platform. The ESP32 will then read both the threshold value and the real-time voltage from the potentiometer. Based on whether the potentiometer voltage exceeds the set threshold, the ESP32 will control the brightness of the LED.

Procedure

- Install latest version of ESP32 library in the Arduino IDE.



- Create a dashboard with widgets as shown below:



- Set the value of the threshold to 1.6 volts and then adjust the potentiometer while observing the changes to the LED brightness.

Delivery Policy

- You should work in groups of four members.
- Each group must send a 20-second video showing the threshold set on the cloud side by side to the adjustment of potentiometer and the effect on LED brightness.
- Each group must submit the solution to the theoretical part in the report.
- Each group must submit the list of files compiled and the storage footprint in the report.
- You should submit a report showing the challenges you faced (if any) and the schematic diagram.
- You should cite any additional resources you used.
- Further details for the submission instructions will be posted later on MS Teams.

Good Luck